

SUPSI

University of Applied Sciences and Arts of Southern Switzerland

Designing With

**A New Educational Module to Integrate
Artificial Intelligence, Machine Learning and
Data Visualization in Design Curricula**

+

Didactic Guidelines

How to cite
Botta, M., Autuori, A., Subet, M., Terenghi, G., (2024). Designing With: A New Educational Module to Integrate Artificial Intelligence, Machine Learning and Data Visualization in Design Curricula. Didactic Guidelines. designingwithai.ch
doi.org/10.5281/zenodo.11355679

Copyright notice
This work is licensed under the Creative Commons Attribution 4.0 (BY) license, which means that the text may be remixed, transformed and built upon and be copied and redistributed in any medium or format even commercially, provided credit is given to the author. <https://creativecommons.org/licenses/by/4.0/>

Creative Commons license terms for re-use do not apply to any content (such as graphs, figures, photos, excerpts, etc.) not original to the Open Access publication and further permission may be required from the rights holder. The obligation to research and clear permission lies solely with the party re-using the material.

First published in May 2024
© Botta, M., Autuori, A., Subet, M., Terenghi, G.



Supported by Movetia. Exchange and mobility.

Contents

1. Didactic Guidelines: Syntax and Methodology	5
1.1 Categories of Objectives	6
1.2 Categories of Knowledge Dimension	7
1.3 Categories of Cognitive Process Dimension	8
1.4 Building the Educational Objectives (EO) and the Instructional Activities	10
1.5 The Framework of Bloom’s Taxonomy Table	11
2. Didactic Guidelines in Practice	15
2.1 Teaching Activity Infrastructure	16
3. Module 1: Getting Familiar	19
3.1 Sub-module 1: Getting Familiar with Machine Learning (ML)	22
3.2 Sub-module 2: Getting Familiar with Generative Artificial Intelligence (AI)	24
3.3 Sub-module 3: Getting Familiar with Data Visualization (DV)	26
4. Module 2: Get in-depth	29
5. Conclusion	33
6. References	34

3.3 Sub-module 2: Getting familiar with Data Visualisation (DV)

This sub-module teaches students to apply Computer Vision (CV) algorithms to analyse and visualise data {DV-Obj}. Students develop procedural knowledge through practical guided activities and they are asked to apply the acquired procedure to a familiar task (e.g. creating an infographic). Students are first provided with a basic literacy of CV algorithms and DV techniques including technical vocabulary {DV-Act1} and primary functionalities {DV-Act2}. This activity is meant to promote factual and conceptual knowledge by teaching the basic elements of the disciplines (CV and DV) and their interrelationship in analysing and visualising complex phenomena (Omena, 2021). Once basic literacy is introduced, specific case studies are provided to contextualise the application of CV within the design research {DV-Act3}. At this point, the teaching moves from a theoretical to a procedural level.

In this direction, students are first introduced to a guided tutorial {DV-Act4.3} on how to analyse data with CV algorithms (e.g. Meme Spector) and visualise the results (e.g. Gephi) starting from setting a research question (DV-Act4.1) and collecting and organising data related to the phenomena {DV-Act4.2}. Then, to familiarise themselves with the procedure, students individually apply what they learned by investigating the dataset to answer the research question {DV-Act5}. Lastly, moving the focus to meta-cognitive knowledge, students are asked to document the process by breaking down the steps and selecting the relevant results {DV-Act6}. This activity aims to create awareness and knowledge of cognition. To foster participation and sharing of results, students are asked to present their work and question others {DV-Act7} (Fig. 9).

Educational Objectives

{DV-Obj}

Students will learn to

apply Computer Vision algorithms

for analyse and visualise data

Instructional Activities

{DV-Act.1}

Activity intended to

provide students

with vocabulary knowledge of CV and DV

{DV-Act.2}

Activity intended to

provide students

with basic functionality of CV algorithms and DV techniques

{DV-Act.3}

Activity intended to

provide students

with practical examples of CV applied to design research

{DV-Act.4}

Activity intended to

explain

the procedure of analysing data with CV algorithms

{DV-Act.4.1}

Activity intended to

introduce students

to familiarising, collecting or creating prompts (data)

{DV-Act.4.2}

Activity intended to

set or align

with a research question

{DV-Act.4.3}

Activity intended to

provide students

a step-by-step tool tutorial, such as Memespector, Gephi

{DV-Act.5}

Activity intended to

allow students

to individually analyse data with computer vision for a research goal

{DV-Act.6}

Activity intended to

provide students

with a structured template for process and results documentation

{DV-Act.7}

Activity intended to

make students

explaining and questioning the process

Sub-module 3: Getting familiar with DV	Cognitive process dimension					
	(1) Remember	(2) Understand	(3) Apply	(4) Analyze	(5) Evaluate	(6) Create
	(a) Factual Knowledge	{DV-Act.1}				
	(b) Conceptual Knowledge	{DV-Act.2} {DV-Act.3}				
	(c) Procedural Knowledge	{DV-Act.4} {DV-Act.4.1, 4.2, 4.3}	{DV-Act.5} {DV-Obj}			
(d) Metacognitive Knowledge				{DV-Act.6}	{DV-Act.7}	

Fig. 9 Sub-Module 1: ‘Getting Familiar’ with Data Visualisation. Educational Objectives and Instructional Activities arranged into the Bloom Taxonomy Table.

6. References

Omena, J.J. (2021). Digital Methods and Technicity-of-the-Mediums. From Regimes of Functioning to Digital Research [Doctoral Dissertation, Universidade Nova de Lisboa]. <http://hdl.handle.net/10362/127961>

Krathwohl, D. R., & Payne, D. A. (1971). Defining and assessing educational objectives. In R. L. Thomdike (Ed.), Educational measurement (pp. 17-45). Washington, DC: American Council on Education.

Anderson, L.W., & Krathwohl, D.R. (2001). A taxonomy for learning, teaching, and assessing abridged edition. Boston: Allyn & Bacon. In The University of Utha, Bloom’s Taxonomy, Revised for 21st-Century Learners.

Designing With: A New Educational Module to Integrate Artificial Intelligence, Machine Learning and Data Visualization in Design Curricula.
designingwithai.ch

Project Coordination
SUPSI - University of Applied Sciences and Arts of Southern Switzerland
Massimo Botta - Project Coordinator
Antonella Autuori - Researcher
Matteo Subet - Researcher
Ginevra Terenghi - Researcher
Desirée Veschetti - Website Designer

Project Partners
NOVA - Universidade NOVA de Lisboa
Janna Joceli Omena - Researcher and Teacher
Eduardo Leite - Researcher and Teacher

EPFL - École Polytechnique Fédérale de Lausanne
Prof. Dr. Jeffrey Huang
Kim Frederick Chando - Researcher and Teacher
Mikhael Johanes - Researcher
Ugo Jean Mario Balducci – Web Developer
Gregory Malyk – Web Developer

